

OLIVIA WANG

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EDUCATION

California Institute of Technology Pasadena, CA | GPA: 4.0/4.0

Jun 2027

B.S. in Computer Science

- Courses: Deep Learning, Diffusion, Operating Systems, Algorithms, Probability, Law & Finance for Startups

EXPERIENCE

Software Development Intern | Amazon Music Culver City, CA

Jun 2026 - Present

- Engineering features for customer onboarding and on-demand listening experiences across mobile and distributed backend systems.
- Utilize AWS services and service-oriented architectures for large-scale applications.

Machine Learning Researcher | Perona Vision Lab, Caltech

Pasadena, CA

Jan 2025 - Present

- Architected PyTorch-based infrastructure to benchmark vision-language diffusion, autoregressive models against human reasoning datasets across 10+ visual tasks.
- Developed a high-precision human-study application for 100 users using Python & Javascript for web-based data collection with precise stimuli timing.
- Engineered Python backends to optimized GPU system performance for batch processing, distributed data parallelization, and concurrent execution on HPC.

Software Engineering Intern | Eli Lilly's Protomer Technologies

Pasadena, CA

Jan 2025 - Jun 2026

- Architected and deployed a full-stack web application using Python (Dash) to visualize high-frequency glucose sensor data, enabling hardware-in-the-loop experiments for critical medical R&D.
- Built a multi-threaded backend in Python to ingest BLE sensor data and deliver real-time ML model predictions for automated insulin regulation logic.

PUBLICATIONS

Wang, O., et al. "[Vision Language Model Comparison with Humans]." Submitted to ECCV 2026.

Wang, O., Larsson, E., Murray, R. "A DNA Part Library for *X. griffithiae*." *ACS Synthetic Biology*. (2025).

PROJECTS

Spectrally Consistent Cloud Removal via NAMM-EDM | PyTorch, SLURM, CUDA

Apr 2026 - Jun 2026

- Achieve state of art metrics SAM of 5.008 and PSNR of 32.43 dB across 13 competing methods with Neural Approximate Mirror Map integrated with EDM diffusion model to enforce spectral consistency.
- Engineered a 3-phase training pipeline on HPC with a custom constraint loss to preserve physical consistency in generated Sentinel-2 imagery.

Sheet Engine with Automated Logic Validation | Python, Pytest, Unittest, API

Jan 2026 - March 2026

- Led a 3-person team to engineer a spreadsheet calculation engine with complex formula parsing and dependency tracking, optimizing for speed and scale.
- Developed automated unit & integrated testing suite, ensuring 100% calculation accuracy and system reliability across edge cases.

SKILLS

- **Software**: Python, Java, C, Javascript, Git, Linux, Containerization, React, Automated Testing
- **Machine Learning**: PyTorch, Tensorflow, scikit-learn, Deep Learning, Computer Vision, LLM, NLP, AWS

LEADERSHIP

- Machine Learning Teaching Assistant for 100+ students, Caltech
- Health Advocate, Social Media & Admissions Ambassador, Caltech
- Team USA and National Figure Skater, Los Angeles Ice Theater

Jan 2026 - Present

May 2024 - Present

Sept 2018 - Jun 2024